Additional Ex (The solutions are posted as in the order in the HW file)

6.3.13

We are asking for the asymptotic distribution of $g(\bar{X}_n)$, where $g(x) = x^3$. The distribution of $\bar{X}_n$ is normal with mean θ and variance $\frac{\sigma^2}{n}$. According to the delta method, the asymptotic distribution of $g(\bar{X}_n)$ should be the normal distribution with mean $g(\theta) = \theta^3$ and variance $\frac{\sigma^2}{n}[g'(\theta)]^2 = \frac{9\theta^4\sigma^2}{n}$.

6.3.15

(a) Clearly, $Y_n \le y$ if and only if $X_i \le y$ for $i = 1, \ldots, n$. Hence,

$$\Pr(Y_n \le y) = \Pr(X_1 \le y)^n = \begin{cases} \left(\frac{y}{\theta}\right)^n, & 0 < y < \theta, \\ 0, & y \le 0, \\ 1, & y \ge \theta. \end{cases}$$

(b) The c.d.f. of Z_n is, for $z < 0$,

$$\Pr(Z_n \le z) = \Pr(Y_n \le \theta + z/n) = (1 + z/[n\theta])^n. \tag{S.6.9}$$

Since $Z_n \le 0$, the c.d.f. is 1 for $z \ge 0$. According to Theorem 5.3.3, the expression in (S.6.9) converges to $\exp(z/\theta)$.

(c) Let $a(y) = y^2$. Then $a'(y) = 2y$. We have $n(Y_n - \theta)$ converging in distribution to the c.d.f. in part (b). The delta method says that, for $\theta > 0$, $n(Y_n^2 - \theta^2)/[2\theta]$ converges in distribution to

7.5.3

The likelihood function for the given sample is $p^{58}(1-p)^{12}$. Among all values of p in the interval $1/2 \le p \le 2/3$, this function is a maximum when $p = 2/3$.

7.5.5a

Let y denote the sum of the observed values $x_1, \ldots, x_n$. Then the likelihood function is

$$f_n(\mathbf{x} \mid \theta) = \frac{\exp(-n\theta)\theta^y}{\prod_{i=1}^n (x_i!)}.$$

(a) If $y > 0$ and we let $L(\theta) = \log f_n(\mathbf{x} \mid \theta)$, then

$$\frac{\partial}{\partial\theta} L(\theta) = -n + \frac{y}{\theta}.$$

The maximum of $L(\theta)$ will be attained at the value of θ for which this derivative is equal to 0. In this way, we find that $\hat{\theta} = y/n = \bar{x}_n$.

(b) If $y = 0$, then $f_n(\mathbf{x} \mid \theta)$ is a decreasing function of θ. Since $\theta = 0$ is not a value in the parameter space, there is no MLE.

7.5.7

Let y denote the sum of the observed values $x_1, \ldots, x_n$. Then the likelihood function is

$$f_n(x \mid \beta) = \beta^n \exp(-\beta y).$$

If we let $L(\beta) = \log f_n(x \mid \beta)$, then

$$\frac{\partial L(\beta)}{\partial \beta} = \frac{n}{\beta} - y.$$

The maximum of $L(\beta)$ will be attained at the value of β for which this derivative is equal to 0. Therefore,

7.5.9

If $0 < x_i < 1$ for $i = 1, \ldots, n$, then the likelihood function will be as follows:

$$f_n(x \mid \theta) = \theta^n \left(\prod_{i=1}^{n} x_i \right)^{\theta - 1}.$$

If we let $L(\theta) = \log f_n(x \mid \theta)$, then

$$\frac{\partial}{\partial \theta} L(\theta) = \frac{n}{\theta} + \sum_{i=1}^{n} \log x_i.$$

Therefore, $\hat{\theta} = -n \Big/ \sum_{i=1}^{n} \log x_i$. It should be noted that $\hat{\theta} > 0$.

7.6.3

The median of an exponential distribution with parameter β is the number m such that

$$\int_0^m \beta \exp(-\beta x) dx = \frac{1}{2}.$$

Therefore, $m = (\log 2)/\beta$, and it follows that $\hat{m} = (\log 2)/\hat{\beta}$. It was shown in Exercise 7 of Sec. 7.5 that $\hat{\beta} = 1/\bar{X}_n$.

7.6.19

The mean of an exponential random variable with parameter β is $1/\beta$, so the method of moments estimator is one over the sample mean, which is also the M.L.E.

7.6.21

The M.L.E. of the mean is the sample mean, which is the method of moments estimator. The M.L.E. of σ^2 is the mean of the X_i^2's minus the square of the sample mean, which is also the method of moments estimator of the variance.

2)

4.6.10

$$\begin{aligned}\mathrm{Var}(X+Y) &= \mathrm{Var}(X)+\mathrm{Var}(Y)+2\mathrm{Cov}(X,Y),\\ \mathrm{Var}(X-Y) &= \mathrm{Var}(X)+\mathrm{Var}(Y)-2\mathrm{Cov}(X,Y).\end{aligned}$$

Since $\mathrm{Cov}(X,Y)<0$, it follows that

$$\mathrm{Var}(X+Y)<\mathrm{Var}(X-Y).$$

4.6.14

(a)

$$\begin{aligned}\mathrm{Var}(X+Y+Z) &= \mathrm{Var}(X)+\mathrm{Var}(Y)+\mathrm{Var}(Z)\\ &\quad +2\mathrm{Cov}(X,Y)+2\mathrm{Cov}(X,Z)+2\mathrm{Cov}(Y,Z)\\ &= 17.\end{aligned}$$

(b)

$$\begin{aligned}\mathrm{Var}(3X-Y-2Z+1) &= 9\mathrm{Var}(X)+\mathrm{Var}(Y)+4\mathrm{Var}(Z)\\ &\quad -6\mathrm{Cov}(X,Y)-12\mathrm{Cov}(X,Z)+4\mathrm{Cov}(Y,Z)\\ &= 59.\end{aligned}$$

4.6.13:

(a) Determine $\mathrm{Var}(X+Y)$

The formula for the variance of the sum of two variables is:

$$\mathrm{Var}(X+Y)=\mathrm{Var}(X)+\mathrm{Var}(Y)+2\mathrm{Cov}(X,Y)$$

Plugging in the values:

$$\mathrm{Var}(X+Y)=9+4+2(-1)$$

$$\mathrm{Var}(X+Y)=13-2=\mathbf{11}$$

(b) Determine $\mathrm{Var}(X - 3Y + 4)$

To solve this, we use two key properties of variance:

- **Constants:** $\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$ (adding a constant like $+4$ doesn't change the spread/variance).
- **Linear Combinations:** $\mathrm{Var}(aX + bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 2ab\mathrm{Cov}(X, Y)$.

For our expression $X - 3Y + 4$, we treat $a = 1$ and $b = -3$:

$$\mathrm{Var}(1X - 3Y + 4) = (1)^2\mathrm{Var}(X) + (-3)^2\mathrm{Var}(Y) + 2(1)(-3)\mathrm{Cov}(X, Y)$$

Plugging in the values:

$$\mathrm{Var}(X - 3Y + 4) = 1(9) + 9(4) + (-6)(-1)$$

$$\mathrm{Var}(X - 3Y + 4) = 9 + 36 + 6 = \mathbf{51}$$

6.1.2

The distribution of $\overline{X}_n$ is (by Corollary 5.6.2) the normal distribution with mean μ and variance σ^2/n. By Theorem 5.6.6,

$$\begin{aligned}\Pr(|\overline{X}_n - \mu| \le c) &= \Pr(\overline{X}_n \le \mu + c) - \Pr(\overline{X}_n \le \mu - c) \\ &= \Phi\left(\frac{c}{\sigma/n^{1/2}}\right) - \Phi\left(\frac{-c}{\sigma/n^{1/2}}\right). \quad \text{(S.6.1)}\end{aligned}$$

As $n \to \infty$, $\frac{c}{\sigma/n^{1/2}} \to \infty$ and $\frac{-c}{\sigma/n^{1/2}} \to -\infty$.
It follows from Property 3.3.2 of all c.d.f.'s that (S.6.1) goes to 1 as $n \to \infty$.

6.2.1

1. Let $\epsilon > 0$. We need to show that

$$\lim_{n\to\infty} \Pr(|X_n - 0| \ge \epsilon) = 0. \quad \text{(S.6.2)}$$

Since $X_n \ge 0$, we have $|X_n - 0| \ge \epsilon$ if and only if $X_n \ge \epsilon$. By the Markov inequality, $\Pr(X_n \ge \epsilon) \le \mu_n/\epsilon$. Since $\lim_{n\to\infty} \mu_n = 0$, Eq. (S.6.2) holds.

6.2.3

By the Chebyshev inequality,

$$\mathrm{Var}(X) \ge 9\Pr(|X - \mu| \ge 3) = 9\Pr(X \le 7 \text{ or } X \ge 13) = 9(0.2 + 0.3) = \frac{9}{2}.$$

6.2.5

By the Chebyshev inequality,

$$\Pr(|X_n - \mu| \le 2\sigma) \ge 1 - \frac{1}{4n}.$$

Therefore, we must have $1 - \frac{1}{4n} \ge 0.99$, or $n \ge 25$.

6.2.7

By the Markov inequality,

$$\Pr(|X - \mu| \ge t) = \Pr(|X - \mu|^4 \ge t^4) \le \frac{\mathbb{E}(|X - \mu|^4)}{t^4} = \frac{\beta_4}{t^4}.$$

Prove Th'm 5.7.3 from the textbook using the change of variables given in its proof.

Theorem 5.7.3: For each $\alpha > 0$ and each $\beta > 0$,

$$\int_0^\infty x^{\alpha-1}e^{-\beta x}\,dx = \frac{\Gamma(\alpha)}{\beta^\alpha}.$$

$$\text{(substituting } x = \frac{y}{\beta},\ dx = \frac{1}{\beta}dy)$$

$$= \frac{1}{\beta}\int_0^\infty \left(\frac{y}{\beta}\right)^{\alpha-1} e^{-y}\,dy$$

$$= \frac{1}{\beta^\alpha}\int_0^\infty y^{\alpha-1}e^{-y}\,dy$$

$$\text{Since } \Gamma(\alpha) = \int_0^\infty y^{\alpha-1}e^{-y}\,dy$$

$$= \frac{\Gamma(\alpha)}{\beta^\alpha}$$

$$\therefore \int_0^\infty x^{\alpha-1}e^{-\beta x}\,dx = \frac{\Gamma(\alpha)}{\beta^\alpha}.$$

Using the definition of expected value, compute $\mathbb{E}(X^3)$ where X has a Gamma distribution. (Hint: See the proof of Theorem 5.7.5 and your result in part (a).)

$$\mathbb{E}[X^3] \quad \text{where } X \sim \text{Gamma}(\alpha, \beta)$$

$$= \int_0^\infty x^3 \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1}e^{-\beta x}\,dx$$

$$= \int_0^\infty \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha+3-1}e^{-\beta x}\,dx$$

$$= \frac{\beta^\alpha}{\Gamma(\alpha)}\int_0^\infty x^{\alpha+3-1}e^{-\beta x}\,dx$$

$$= \frac{\beta^\alpha}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha+3)}{\beta^{\alpha+3}}$$

$$= \frac{\Gamma(\alpha+3)}{\beta^3\Gamma(\alpha)}.$$

For each case below, compute the moment generating function (MGF) of $X_1 + X_2 + X_3 + X_4 + X_5$ and identify its distribution. Assume that X_i, $i = 1, \ldots, 5$ are independent.

Since MGFs factor over independent variables:

$$\mathbb{E}[e^{t(X_1+X_2+X_3+X_4+X_5)}] = \prod_{i=1}^{5} \mathbb{E}[e^{tX_i}].$$

Thus, we compute the MGF for a single X_i in each case.

- **Bernoulli**(p)

$$\mathbb{E}[e^{tX}] = (1-p) + pe^t.$$
$$\mathbb{E}[e^{t(X_1+\cdots+X_5)}] = (1-p+pe^t)^5.$$
$$\mathbb{E}[e^{t(X_1+\cdots+X_5)}] = \textbf{Binomial}(5, p) \tag{1}$$

- **Binomial**(n_i, p)

$$\mathbb{E}[e^{tX}] = (1-p+pe^t)^n.$$
$$\mathbb{E}[e^{t(X_1+\cdots+X_5)}] = (1-p+pe^t)^{\sum n_i}.$$
$$\mathbb{E}[e^{t(X_1+\cdots+X_5)}] = \textbf{Binomial}(\sum_{i=1}^{5} n_i, p) \tag{2}$$

- **Poisson**(λ_i)

$$\mathbb{E}[e^{tX}] = e^{\lambda(e^t-1)}.$$
$$\mathbb{E}[e^{t(X_1+\cdots+X_5)}] = e^{(e^t-1)\sum \lambda_i}.$$
$$\mathbb{E}[e^{t(X_1+\cdots+X_5)}] = \textbf{Poisson}(\sum_{i=1}^{5} \lambda_i)$$

- **Normal**(μ_i, σ_i^2)

$$\mathbb{E}[e^{tX}] = e^{\mu t + \frac{1}{2}\sigma^2 t^2}.$$
$$\mathbb{E}[e^{t(X_1+\cdots+X_5)}] = e^{\sum(\mu_i t + \frac{1}{2}\sigma_i^2 t^2)}.$$
$$\mathbb{E}[e^{t(X_1+\cdots+X_5)}] = \textbf{Normal}(\sum_{i=1}^{5} \mu_i, \sum_{i=1}^{5} \sigma_i^2)$$

- **Gamma**(α_i, β)

$$\mathbb{E}[e^{tX}] = (1-t\beta^{-1})^{-\alpha}, \quad \text{for } t < \frac{1}{\beta}.$$
$$\mathbb{E}[e^{t(X_1+\cdots+X_5)}] = (1-t\beta^{-1})^{\sum -\alpha_i}.$$
$$\mathbb{E}[e^{t(X_1+\cdots+X_5)}] = \textbf{Gamma}(\sum_{i=1}^{5} \alpha_i, \beta)$$

- **Exponential**(β)

$$\mathbb{E}[e^{tX}] = (1-t\beta^{-1})^{-1}, \quad \text{for } t < \beta.$$
$$\mathbb{E}[e^{t(X_1+\cdots+X_5)}] = (1-t\beta^{-1})^{-5}.$$
$$\mathbb{E}[e^{t(X_1+\cdots+X_5)}] = \textbf{Gamma}(5, \beta)$$

7.7.1

The joint p.f. is

$$f_n(\boldsymbol{x}|p) = p^t(1-p)^{n-t}.$$

8.2.9

It is known that $\overline{X}_n$ has the normal distribution with mean μ and variance σ^2/n. Therefore, $(\overline{X}_n - \mu)/(\sigma/\sqrt{n})$ has a standard normal distribution and the square of this variable has the χ^2 distribution with one degree of freedom.

8.4.3

$X_1 + X_2$ has the normal distribution with mean 0 and variance 2. Therefore, $Y = (X_1 + X_2)/\sqrt{2}$ has a standard normal distribution. Also, $Z = X_3^2 + X_4^2 + X_5^2$ has the χ^2 distribution with 3 degrees of freedom, and Y and Z are independent. Therefore, $U = \frac{Y}{(Z/3)^{1/2}}$ has the t distribution with 3 degrees of freedom. Thus, if we choose $c = \sqrt{3/2}$, the given random variable will be equal to U.

Exercise: Let Z_1, Z_2, Z_3 be i.i.d $N(0,1)$ random variables. Consider the vector (Z_1, Z_2, Z_3) as a random point in 3-dimensional space. Let R be the length of the radius vector connecting this point to the origin. Find the density functions of:

(a) R

We know that $Z_1^2 + Z_2^2 + Z_3^2 \sim \chi_3^2 \sim \text{Gamma}(3/2, 1/2)$, so $R^2 \sim \text{Gamma}(3/2, 1/2)$. Using the transformation method, if $R = \sqrt{Y}$ then $f_R(r) = f_Y(r^2) \cdot |2r|$. The density of $Y \sim \text{Gamma}(3/2, 1/2)$ is: $f_Y(y) = \frac{(1/2)^{3/2}}{\Gamma(3/2)} y^{3/2-1} e^{-y/2} = \frac{1}{2\sqrt{2}\Gamma(3/2)} y^{1/2} e^{-y/2}$. So

$$f_R(r) = f_Y(r^2) \cdot |2r| = \frac{(1/2)^{3/2}}{\Gamma(3/2)} r^{3-1} e^{-r^2/2} \text{ for } r > 0$$

(b) R^2

$$R^2 = Z_x^2 + Z_y^2 + Z_z^2 = \chi_3^2$$